Minghao Fu

Ph.D. Student · Halıcıoğlu Data Science Institute

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Education

University of California, San Diego — *San Diego, CA, USA*
Ph.D. Student, Halıcıoğlu Data Science Institute — *Aug 2025 – Present*
- Advisor: Biwei Huang

Carnegie Mellon University — *Pittsburgh, PA, USA*
Visiting Student, CLeaR, Philosophy Department — *Jun 2024 – Nov 2024*
- Advisors: Peter Spirtes & Kun Zhang

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — *Abu Dhabi, UAE*
M.S. in Machine Learning — *Aug 2023 – Jun 2025*
- Advisor: Kun Zhang

University of Electronic Science and Technology of China — *Chengdu, China*
B.S. in Software Engineering — *Sep 2019 – Jun 2023*
- GPA 3.83/4.0. Graduated with Honor Research, Outstanding Undergraduate Thesis

Research Interests

Robot learning, world models, planning, representation learning, and reinforcement learning.

Publications & Preprint

World Model & Representation Learning

LeWAM: End-to-End World and Action Modeling with JEPAs.
***Minghao Fu**[*], Tavis Siebert[*], Eryk Halicki, Randall Balestriero.*
Under Review. [Paper] [Website] [Code]

Back to Parsimonious Latents: Learning Task-Centric World Models from Visual Foundations.
***Minghao Fu**, Fan Feng, Nicklas Hansen, Biwei Huang.*
NeurIPS 2026. [Paper] [Website] [Code]

Factored Generative Models through Mechanism Diversity.
***Minghao Fu**, Selena Ge, Hongjia Liu, Zijian Li, Fan Feng, Kun Zhang, Biwei Huang.*
NeurIPS 2026.

RSIGame: Autonomous Agentic Game Development with Recursive Self-improvement.
*Wenyi Wu[*], **Minghao Fu**[*], Jieyu You, Kun Zhou, Siqi Liu, Aayush Salvi, Yiheng Lin, Ce Zhang, Xiaohan Lan, Jiahui Zhu, Yujie Zhong, Qi She, Biwei Huang.*
Under Review. [Paper] [Website] [Code] [Data]

Video2World: Benchmarking Coding Agents for Interactive World Modeling from Embodied Videos.
*Jinzhou Tang, Zijun Zhang, Jing Yang, Yuchen Yan, Kun Zhou, Lingjun Mao, Ruobing Han, Jinglin Cao, Wenpeng Xu, Lukun He, **Minghao Fu**, Fan Feng, Biwei Huang.*
Under Review. [Paper] [Website] [Code] [Data]

Learning Task-Sufficient World Models by Synergizing Agentic Exploration and Structured Modeling.
*Fan Feng[*], Yujia Zheng[*], **Minghao Fu**, Yongqiang Chen, Guangyi Chen, Kevin Murphy, Biwei Huang, Kun Zhang.*
ICML 2026. [Paper]

DreamSAC: Learning Hamiltonian World Models via Symmetry Exploration.
*Jinzhou Tang, Fan Feng, **Minghao Fu**, Wenjun Lin, Jing Yang, Biwei Huang, Keze Wang.*
CVPR 2026. [Paper] [Code]

Ada-Diffuser: Latent-Aware Adaptive Diffusion for Decision Making.
*Fan Feng, Selena Ge, **Minghao Fu**, Zijian Li, Yujia Zheng, Zeyu Tang, Yingyao Hu, Biwei Huang, Kun Zhang.*
ICLR 2026. [Paper] [Website]

Theory of Representation Learning

PersonaX: Multimodal Datasets with LLM-Inferred Behavior Traits.
*Loka Li[*], Wong Yu Kang[*], **Minghao Fu**, Guangyi Chen, Zhenhao Chen, Gongxu Luo, Yuewen Sun, Salman Khan, Peter Spirtes, Kun Zhang.*
ICLR 2026. [Paper] [Website] [Code] [Data]

Learning General Causal Structures with Hidden Dynamic Process for Climate Analysis.
***Minghao Fu**, Biwei Huang, Zijian Li, Yujia Zheng, Ignavier Ng, Guangyi Chen, Yingyao Hu, Kun Zhang.*
ICML 2026. [Paper]

Towards Identifiability of Hierarchical Temporal Causal Representation Learning.
*Zijian Li[*], **Minghao Fu**[*], Junxian Huang, Yifan Shen, Ruichu Cai, Yuewen Sun, Guangyi Chen, Kun Zhang.*
NeurIPS 2025. [Paper] [Code]

Online Time Series Forecasting with Theoretical Guarantees.
*Zijian Li, Changze Zhou, **Minghao Fu**, Sanjay Manjunath, Fan Feng, Guangyi Chen, Yingyao Hu, Ruichu Cai, Kun Zhang.*
NeurIPS 2025. [Paper] [Code]

Professional Experience

ByteDance *San Jose, CA, USA*
Research Scientist Intern, Algorithm and Foundation Team *Jun 2026 – Sep 2026*

- Mentor: Yujie Zhong
- Used coding agents to build interactive worlds in a self-improving way; this line of work became *RSIGame*.
- Built the coding-agent RL training infrastructure behind a production system, together with its data processing pipeline.

Shanghai Artificial Intelligence Laboratory *Shanghai, China*
Research Intern, Ark NLP Group *Nov 2022 – Mar 2023*

- Mentor: Lingpeng Kong
- Research Topics: Non-Autoregressive Transformer, Linear Attention, Machine Translation

Awards, Fellowships, & Honors

2026	**Silver Reviewer**, ICML
Dec 2025	**Volunteer Program, registration award**, NeurIPS
Jun 2025	**Machine Learning Citizen Award (1/37)**, MBZUAI
May 2023	**The First-ever Human Phenotype Project Hackathon, Rank-1st**, Weizmann Institute of Science & MBZUAI
May 2023	**Advanced Study Scholarship (< 5%)**, UESTC
Apr 2023	**Undergraduate High-Level Paper Award (<1%)**, UESTC
Jun 2023	**Honor Research Scholarship (< 1%)**, UESTC
2020-2023	**Excellent Student Scholarship (<10%)**, UESTC

Service & Professional Activities

Service & Outreach Volunteer, Neurips 2025 Dec 2025
Mentor, "Sadeeq" MBZUAI Buddy Aug 2024 – Jun 2025
Mentor, MBZUAI Undergraduate Research Internship Program (UGRIP) Jun 2024 – Aug 2024

Peer Review

Journals: TMLR, IEEE TIP
Conferences: ICML, ICLR, NeurIPS, CVPR, AAAI, BMVC

Skills

Deep Learning: PyTorch, Tensorflow, scikit-learn, Weights & Biases
Coding Agents: Claude Code, Codex, MCP tooling, agent harness design
Programming: Python, C++, Bash, MATLAB